

LECTURE 8

6. RINGS AND FIELDS

6.1. Introduction to rings. Inspired by the fact that we have seen that some laws hold for the integers, the complex numbers, matrices, polynomials, etc, we have the following definition:

Definition 6.1. A set R together with two binary operations $+, \cdot : R \times R \rightarrow R$ is called a *ring* if the following holds:

- (A1) $a + b \in R$ for all $a, b \in R$ (closure of addition)
- (A2) $a + (b + c) = (a + b) + c$ for all $a, b, c \in R$ (associativity of addition)
- (A3) $a + b = b + a$ for all $a, b \in R$ (commutativity of addition)
- (A4) there exists $0 \in R$ such that $a + 0 = a$ for all $a \in R$ (existence of zero)
- (A5) for every $a \in R$ there is an element $-a \in R$ such that $a + (-a) = 0$ (existence of additive inverse)
- (M1) $a \cdot b \in R$ for all $a, b \in R$ (closure of multiplication)
- (M2) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$ (associativity of multiplication)
- (D) $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$ (distributivity)

Example 6.2. We have that $(\mathbb{Z}, +, \cdot)$, with $+$ and $\cdot$ as ordinary addition and multiplication, is a ring. Also $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{C}$ with ordinary addition and multiplication form rings.

Example 6.3. We consider the set $2\mathbb{Z}$ of even integers. This is a subset of the integers $\mathbb{Z}$. We first check that for $a, b \in 2\mathbb{Z}$, we have $a + b$ and $a \cdot b$ still being in $2\mathbb{Z}$. This is true (give explicit details, e.g. let $a = 2m$ and $b = 2n$ for some $m, n \in \mathbb{Z}$, then $a + b = 2(m + n)$, which is clearly in $2\mathbb{Z}$). Now axioms (A2), (A3), (M2), and (D) are inherited from $\mathbb{Z}$. It remains to check that $0 \in 2\mathbb{Z}$ and that the additive inverses of even numbers are again even numbers. As the answer is yes to both cases, we have that $2\mathbb{Z}$ is a ring.

Example 6.4 (Non-example). The set of all odd numbers is not a ring because the sum of two odd numbers is even.

In the above examples of rings we also have that

- (M3) $a \cdot b = b \cdot a$ for all $a, b \in R$ (commutativity of multiplication)

holds. Such rings are called *commutative rings*.

Example 6.5 (Non-example). The set $M_2(\mathbb{R})$ of all 2×2 -matrices with real entries is a ring. The zero element is the matrix $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and the additive inverse of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $\begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix}$. The laws (A1), (A2), (A3), (M1), (M2) and (D) are known from linear algebra. It is straightforward to show that this is a non-commutative ring, for example, taking $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ we have $AB = \begin{pmatrix} 4 & 6 \\ 4 & 6 \end{pmatrix}$ but $BA = \begin{pmatrix} 3 & 3 \\ 7 & 7 \end{pmatrix}$. So $AB \neq BA$.

In the above example, we also have that $M_2(\mathbb{R})$ has so-called zero divisors:

$$\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Definition 6.6. Let R be a ring. An element $a \neq 0$ is called a *zero divisor* if there is an element $b \neq 0$ such that $a \cdot b = 0$.

More examples of zero divisors to follow.

6.2. Rings with unity. Most rings that we will encounter have the property:

(M4) there is an element $1 \in R$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in R$ (existence of unity)

Example 6.7. In $\mathbb{Z}$, $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$, the number 1 is the unity.

Example 6.8 (Non-example). In $2\mathbb{Z}$ there is no unity. Indeed, for $\ell \in \mathbb{Z}$ the equation $(2k)(2\ell) = (2\ell)(2k) = 2\ell$ has no integer solution $k \in \mathbb{Z}$.

Example 6.9. In $M_2(\mathbb{R})$ the unity is $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

6.3. Fields.

Definition 6.10. A commutative ring R with unity such that

(M5) for every $a \neq 0$ in R there is $a^{-1} \in R$ such that $a \cdot a^{-1} = 1$ holds is called a *field*.

Example 6.11. Considering our examples of commutative rings with unity we find that $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$ are fields using $1/a$ as a^{-1} .

Example 6.12 (Non-example). In $\mathbb{Z}$ however $1/a$ is usually not in the ring $\mathbb{Z}$, for example $1/2$ is not an integer. In fact the only invertible elements in $\mathbb{Z}$ are ± 1 . For other $a \in \mathbb{Z}$ we cannot find any $b \in \mathbb{Z}$ with $a \cdot b = 1$. Therefore $\mathbb{Z}$ is not a field.

Example 6.13 (Non-example). The commutative ring $2\mathbb{Z}$ has no unity, so is not a field.

Example 6.14 (Non-example). The ring $M_2(\mathbb{R})$ is not commutative, so is not a field.

6.4. Rings of congruence classes. Let n be an integer ≥ 2 . Recall that two integers a, b are congruent modulo n (written $a \equiv b \pmod{n}$) if n divides $a - b$. Congruence modulo n is an equivalence relation (that is, reflexive: $a \equiv a$, symmetric: $a \equiv b \implies b \equiv a$, and transitive: $a \equiv b$ and $b \equiv c \implies a \equiv c$). Therefore it partitions the integers into disjoint equivalence classes. Denote the class containing a by $[a]$.

We want to make the set of classes $R = \{[0], [1], [2], \dots, [n-1]\}$ into a ring by defining $+$ and $\cdot$.

Let $[a] + [b] = [a + b]$ and $[a] \cdot [b] = [a \cdot b]$ (here the left-hand sides of both equations are the operation that we are defining, where the operations on the right-hand sides are known operations between ordinary numbers).

We need to check that our $+$ and $\cdot$ are well defined, that is that the result is independent of the choice of representatives a, b of the classes involved. If a', b' are other representatives of these classes then $a - a' = k \cdot n$ and $b - b' = \ell \cdot n$ for some integers k, ℓ . Hence

$$[a'] + [b'] \stackrel{\text{by def.}}{=} [a' + b'] = [a - k \cdot n + b - \ell \cdot n] = [a + b + n(-k - \ell)] = [a + b],$$

so $+$ is well defined.

Likewise for $\cdot$ we have

$$[a'] \cdot [b'] \stackrel{\text{by def.}}{=} [a' \cdot b'] = [(a - kn)(b - \ell n)] = [ab + n(k\ell n - bk - a\ell)] = [a \cdot b],$$

showing that this operation is also well defined.

The axioms (A1) - (A5), (M1), (M2) and (D) are now easy to show, since they hold for the integers $\mathbb{Z}$. The ring of congruence classes modulo n is denoted $\mathbb{Z}_n$. It is a commutative ring and has unity $[1]$.

Example 6.15. In $\mathbb{Z}_6$ we have

$$\begin{aligned} [3] + [2] &= [5], & [5] + [4] &= [9] = [3] \\ [4] \cdot [2] &= [8] = [2], & [2] \cdot [3] &= [6] = [0]. \end{aligned}$$

This shows that there are zero divisors in $\mathbb{Z}_6$. A zero divisor is not invertible (exercise: prove this fact), so $\mathbb{Z}_6$ cannot be a field.

Example 6.16. In $\mathbb{Z}_7$ we have

$$\begin{aligned} [1] \cdot [1] &= [1] \\ [2] \cdot [4] &= [8] = [1] \\ [3] \cdot [5] &= [15] = [1] \\ [6] \cdot [6] &= [36] = [1]. \end{aligned}$$

So all non-zero elements in $\mathbb{Z}_7$ are invertible. Therefore $\mathbb{Z}_7$ is a field.

In general $\mathbb{Z}_n$ is a field if n is a prime, but otherwise it is not a field, as we will see later.

Theorem 6.17. Let $n \in \mathbb{Z}_{\geq 2}$. Then $[a] \neq [0]$ has an inverse in $\mathbb{Z}_n$ if and only if $(a, n) = 1$.

Proof. Assume first that $(a, n) = d \geq 2$. Then $a = da'$ and $n = dn'$, for some $a', n' \in \mathbb{N}$, and further

$$[a] \cdot [n'] = [an'] = [da'n'] = [a'n] = [0].$$

So $[a]$ is a zero divisor, which implies that $[a]$ has no inverse.

Assume now for the converse that $(a, n) = 1$. Then using Euclid's algorithm on a and n followed by reverse substitution, we find $b, c \in \mathbb{Z}$ such that $1 = a \cdot b + n \cdot c$. In $\mathbb{Z}_n$ this becomes $[1] = [a] \cdot [b]$. Thus $[b]$ is an inverse of $[a]$. $\square$

Example 6.18. Find the inverses of $[42]$ and $[70]$ in $\mathbb{Z}_{275}$ if they exist.

Solution: First we check if 42 and 70 have common factors with 275:

$$42 = 2 \cdot 3 \cdot 7, \quad 70 = 2 \cdot 5 \cdot 7, \quad 275 = 5^2 \cdot 11$$

Therefore $(42, 275) = 1$ and so $[42]$ has an inverse. However $(70, 275) = 5$, so $[70]$ has no inverse.

We now compute $[42]^{-1}$. Applying Euclid's algorithm gives:

$$\begin{aligned} 275 &= 6 \cdot 42 + 23 \\ 42 &= 2 \cdot 23 - 4 \\ 23 &= 6 \cdot 4 - 1 \end{aligned}$$

so backtracking yields

$$\begin{aligned}
 1 &= 6 \cdot 4 - 23 \\
 &= 6(2 \cdot 23 - 42) - 23 \\
 &= 11 \cdot 23 - 6 \cdot 42 \\
 &= 11(275 - 6 \cdot 42) - 6 \cdot 42 \\
 &= 11 \cdot 275 - 72 \cdot 42.
 \end{aligned}$$

Hence $1 = 11 \cdot 275 - 72 \cdot 42$, which implies that $[1] = [-72][42]$. Therefore $[42]^{-1} = [-72] = [203]$. The answer is then: $[70]$ has no inverse in $\mathbb{Z}_{275}$ and $[42]^{-1} = [203]$.

Theorem 6.19. *The ring $\mathbb{Z}_n$ is a field if and only if n is a prime number.*

Proof. The statement follows immediately from the previous theorem since $(a, p) = 1$ for all $1 \leq a \leq p - 1$ if and only if p is prime. $\square$

Here is another example of a ring.

Example 6.20. Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ where multiplication and addition is defined as it is for the complex numbers. Verify that $\mathbb{Z}[i]$ is a commutative ring and find out which elements in $\mathbb{Z}[i]$ have a multiplicative inverse. Is $\mathbb{Z}[i]$ a field?

Solution: We first justify that $\mathbb{Z}[i]$ is a ring: the set $\mathbb{Z}[i]$ inherits axioms from $\mathbb{C}$, so we only need to check that $0 \in \mathbb{Z}[i]$ (it obviously is) and that if $a + bi \in \mathbb{Z}[i]$ then also its additive inverse $-a - bi \in \mathbb{Z}[i]$ (this is clear too).

Commutativity is also inherited from $\mathbb{C}$.

In $\mathbb{C}$ we can form a multiplicative inverse of $a + bi$ (unless $a = b = 0$) by

$$\frac{1}{a + bi} = \frac{a - bi}{a^2 + b^2} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.$$

This inverse is in $\mathbb{Z}[i]$ if and only if $\frac{a}{a^2 + b^2}$ and $\frac{b}{a^2 + b^2}$ are both integers, which is equivalent to $(a^2 + b^2) \mid a$ and $(a^2 + b^2) \mid b$ with $a + bi \neq 0$.

Case I: Suppose $a = 0$. Then $(a^2 + b^2) \mid a$ is equivalent to $b^2 \mid 0$, which is always true. Next, the condition $(a^2 + b^2) \mid b$ is equivalent to $b^2 \mid b$, which yields $b \in \{-1, 0, 1\}$. Hence $a + bi$ equals $-i$ or i (recall that 0 is excluded).

Case II: Suppose $a \neq 0$. Then $(a^2 + b^2) \mid a$ implies that $a^2 + b^2 \leq |a|$ and therefore $b = 0$, and hence $a \in \{-1, 0, 1\}$. As before, we obtain that $a + bi$ equals ± 1 .

Thus the only invertible elements in $\mathbb{Z}[i]$ are ± 1 and $\pm i$. Hence $\mathbb{Z}[i]$ is not a field.